

SHIVAM KUMAR

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EDUCATION

Indian Institute of Information Technology Bhagalpur

Aug 2024 - Sep 2028

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.05/10

SKILLS

Programming Languages: HTML5, CSS, JavaScript, C, C++

Libraries & Frameworks: React, Next.js, Tailwind CSS

Backend & APIs: Node.js, Express.js, JWT authentication, bcrypt hashing, Tokens generation

Databases: MongoDB, Mongoose

DevOps & Tools: Docker, Nginx, GitHub, Git, Virtual Box, VS Code

Cloud & Deployment: AWS (EC2), Google Cloud (Basic), Cloudflare Workers, Vercel, Netlify

PROJECTS

PORTFOLIO

Jun 2025 - Sep 2025

React | Node.js | Express.js | Tailwind CSS | MongoDB | Google Gemini API | Familiar: Three.js, Framer Motion

- Developed a dynamic, fully responsive personal portfolio from scratch using the MERN stack to showcase my skills in a modern, interactive format.
- Built an interactive web terminal allowing users to navigate the site via custom commands, demonstrating advanced frontend development and a creative user experience.
- Integrated the GitHub API to dynamically fetch and display my latest repositories and contribution activity, providing live proof of continuous coding work.
- Integrated complex 3D animated backgrounds using Three.js and a custom HTML5 Canvas particle system, creating an immersive and visually engaging environment.

Path Finder

React | Node.js | Tailwind CSS | JavaScript (ES6+) | Vite | HTML5 | DSA

- Developed Network Breach Pathfinder, a React.js + Vite pathfinding visualizer featuring a responsive 20x50 interactive grid and drag-and-drop obstacle creation.
- Implemented Dijkstra's Algorithm and A* Search in JavaScript using a min-heap priority queue and Manhattan distance heuristic for efficient shortest-path computation.
- Engineered real-time visualization with optimized node animation, DOM updates, and state management to improve frontend performance and user experience.
- Built modular, production-ready architecture with reusable components, clean algorithm utilities, and maintainable CSS-based UI design.
- Achieved high-performance frontend metrics: LCP (1.83s), CLS (0), INP (32ms), ensuring smooth and responsive user experience

EXPERIENCE

- Designed and developed full-stack web applications using MERN stack with scalable architecture and modular components
- Implemented authentication systems and optimized performance for faster load times (<2s)
- Collaborated with senior developers to design and develop web-based CTF challenges focusing on real-world security vulnerabilities
- Deployed applications using cloud platforms like Vercel, Netlify, and AWS EC2

ACHIEVEMENTS & EXTRACURRICULARS

- Top 10 finalist - Smart India Hackathon (IIT Bhagalpur) as Team lead
- CISCO - Introduction to Cybersecurity
- Volunteer at Unnati Welfare Society
- Top 7% finalist - National Level OWASP CTF
- Advent of Cyber 2024 on TryHackMe